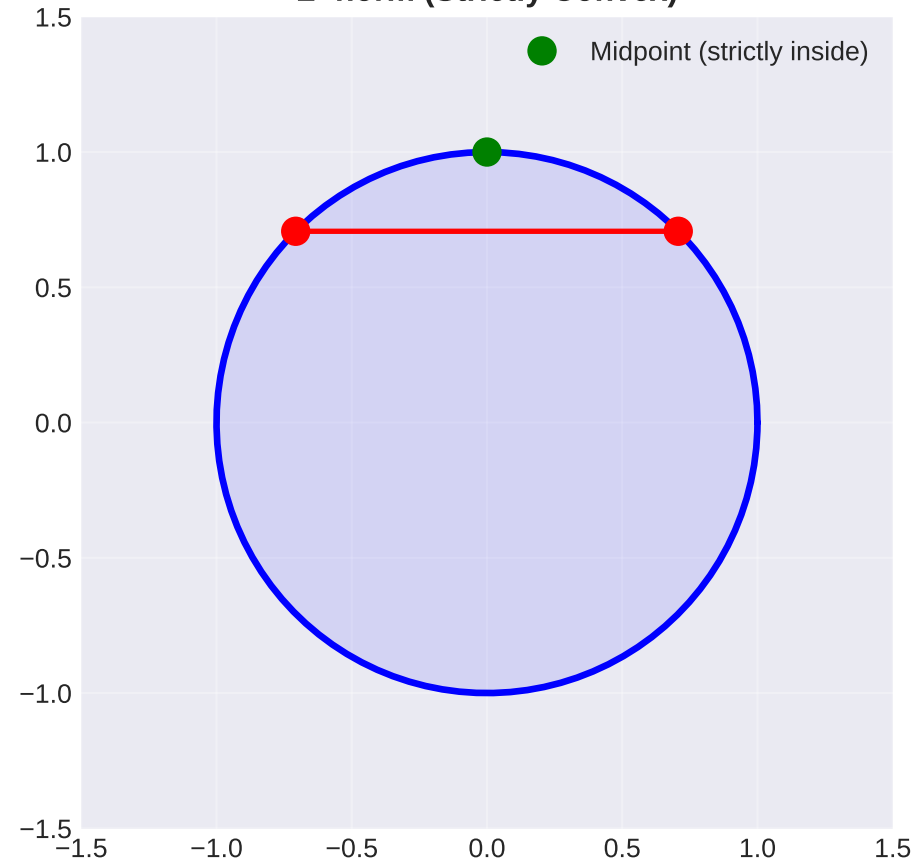
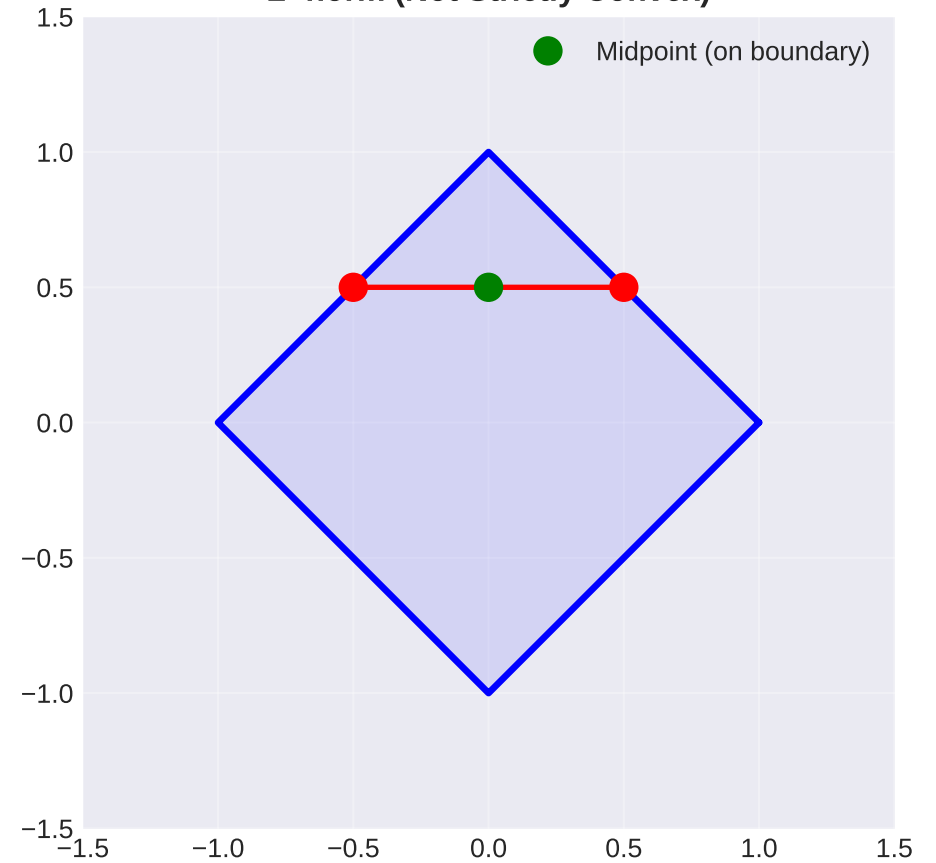


Unit Balls in Different Normed Spaces

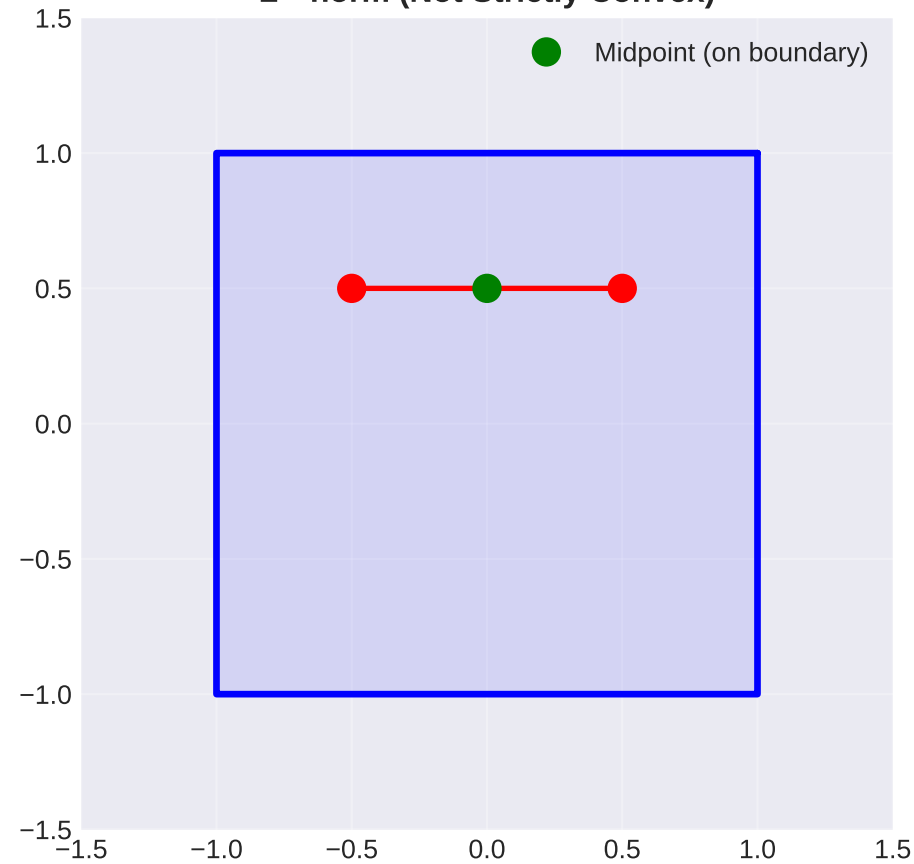
L^2 norm (Strictly Convex)



L^1 norm (Not Strictly Convex)



L^∞ norm (Not Strictly Convex)



$C[0,1]$: Space of continuous functions

Unit ball: $\{f : \|f\|_\infty \leq 1\}$

Example: $f(t) = t$ and $g(t) = -t$
Both have $\|f\|_\infty = \|g\|_\infty = 1$
Midpoint: $(f+g)/2 = 0$ (strictly inside)

Not strictly convex